Hongrui Zhu

blackornate.github.io | github.com/BlackOrnate | linkedin.com/in/hongruizhu

Research Interests

Medical image analysis and computational pathology; pretrained visual representations; multimodal medical AI; language models and agents for healthcare and biomedical research.

Education

Stony Brook University Sep 2024 – May 2026

M.S. in Biomedical Informatics | GPA 3.88/4.00

Thesis advisor: Professor Chao Chen

Dalian Jiaotong University Sep 2019 – Jun 2024

B.Sc. in Mathematics and Applied Mathematics; B.Eng. in Software Engineering

Weighted average grade: 84.58/100

Research Experience

Graduate Researcher in Brain Histopathology Jun 2025 – May 2026

Biomedical Informatics, Stony Brook University | Advisor: Professor Chao Chen

github.com/BlackOrnate/CellSeg-UNICls

- Developed CellSeg-UNICls, combining Cellpose-SAM instance segmentation, frozen UNI2 embeddings, and an MLP classifier. Implemented cell cropping, feature extraction, classification, and prediction export for expert review.
- Adapted the decoupling approach of CellViT++ to pair Cellpose-SAM segmentation with a separate cell classifier after identifying a segmentation bottleneck in joint models.
- Contributed to a dataset of 48 brain histopathology patches from 6 patients with 13,697 annotated instances. Converted expert-labelled points into boundaries using the QuPath SAM extension and refined masks through repeated expert review.
- Reproduced and compared CellViT, CellViT++, PointFormer, and Cellpose-SAM using segmentation, detection, and classification metrics; examined class-specific errors and explored fragment merging for over-segmented instances.

FishLEN Research Project Jan – Jun 2025

Dormant Biology Lab, Stony Brook University | Advisor: Professor Chi-Kuo Hu

github.com/BlackOrnate/FishLEN

- Independently developed a fish-length estimation pipeline from laboratory-provided raw images, including data organization, annotation preparation, model implementation, and evaluation.
- Generated candidate masks with SAM and manually selected fish regions. Combined U-Net with a ResNet34 encoder for segmentation and ResNet18 with an MLP for length regression.

Research Project in Pathology Image Analysis Nov – Dec 2024

Biomedical Informatics, Stony Brook University | Advisor: Professor Chao Chen

- Reviewed cell segmentation and classification literature, reproduced early segmentation baselines including HoVer-Net, and used QuPath for pathology image and annotation processing.

Research Intern Oct – Dec 2023

Center for Innovation in Brain Science, University of Arizona | Remote | Advisor: Professor Rui Chang

- Modernized a melanoma image-analysis and model-training workflow from TensorFlow 1 to TensorFlow 2; debugged training and streamlined redundant computation to improve runtime and GPU memory use.
- Presented progress in twice-weekly Zoom meetings and communicated implementation questions and updates by email. The project also informed the software engineering undergraduate thesis.

Selected Projects

Nuclei Instance Segmentation on PanNuke Spring 2025

AMS 691 Medical Image Analysis, Stony Brook University | Instructor: Dr. Chenyu You

github.com/BlackOrnate/MCV

- Designed, implemented, and optimized the model architecture in a team project. Adapted experience from FishLEN to binary nuclei-mask prediction and HoVer-map regression for instance segmentation.

Medical Question Answering Agent

Independent learning prototype with simulated medical data

github.com/BlackOrnate/langgraph-medical-multi-agent

- Implemented basic LangGraph routing to modules for simulated visit-history lookup through MCP, FAISS retrieval over example guidelines, and Text-to-SQL queries over a mock doctor database; used Ollama for local language models.

AI Research Workspace

Independent learning prototype for research assistance

github.com/BlackOrnate/AI-Research-Workspace

- Built a basic workflow connecting OpenAlex paper search, FAISS retrieval over sample paper material, MCP note lookup, and concept explanation, with routed outputs combined into a response.

Theses

CellSeg-UNICls: Decoupled Segmentation and Embedding-based Cell Classification for Brain Histopathology. M.S. thesis, Stony Brook University, 2026.

Discussion on Methods for Solving Linear Programming Problems. Undergraduate thesis in Mathematics and Applied Mathematics, Dalian Jiaotong University, 2024.

Neural Network Algorithm for Cancer Cell (Melanoma) Detection based on Image Data. Undergraduate thesis in Software Engineering, Dalian Jiaotong University, 2024.

Honors and Awards

- Outstanding Undergraduate Thesis Award, Mathematics and Applied Mathematics, 2024. One of 2 recipients among 67 School of Science graduates, Dalian Jiaotong University.
- Outstanding Undergraduate Thesis Award, Software Engineering, 2024. One of 14 recipients among more than 500 software engineering graduates, Dalian Jiaotong University.
- National Second Prize and Provincial First Prize, computer game competition, 2022. National event: China Collegiate Computer Game Competition and 16th China Computer Game Championship. Implemented, debugged, and optimized game-playing algorithms in a two-student team.
- Second-Class University Scholarship, 2020–2021; Third-Class University Scholarship, 2019–2020 and 2021–2022. Dalian Jiaotong University.

Technical Skills

Primary language: Python. Additional coursework and prior project experience: Java, C, SQL, MATLAB.

Deep learning and scientific computing: PyTorch, TensorFlow, NumPy, Pandas, SciPy, OpenCV, scikit-image, Matplotlib.

Image analysis and development: QuPath, cell segmentation and classification, Git, GitHub, Linux.

Agent tools used in learning projects: LangGraph, LangChain, Ollama, FAISS, MCP, Text-to-SQL.

Technical Writing and Service

- Authored 76 technical blog posts between Jan 2022 and Oct 2024, with 43,614 cumulative views as of Nov 2026.
- Provided remote tutoring in mathematics and science to elementary school children of frontline healthcare workers responding to COVID-19, Feb – Oct 2020.